Hopping on Number Maze and Programming

Grade 6

Explore computer programming by writing steps for solving a maze. A program is basically the set of instructions to tell the computer what to do. To be sure that the program is working and solving the problem, it also needs to be tested. The activity provides a group of students an opportunity to play different roles in the activity and understand the basics of computer programming.



Note for Teachers: Learning Outcomes

- Actively listen to the feedback and check the program (instructions). Communicate when the program isn't correct. *Literacy: Listening, Speaking*
- Give a set of precise instructions that programs an object to move efficiently from one point to another. Computational Thinking - *Abstraction, Algorithmic Thinking*
- Observe patterns in movement and apply them to solve navigation challenges. Computational Thinking – *Pattern Recognition*
- Identify errors in a sequence of instructions and suggest corrections. *Computational Thinking – Debugging, Logical Reasoning*
- Predict the outcome of a set of instructions before testing it. *Computational Thinking – Prediction, Problem Solving*
- Collaborate with peers to test and improve a program. *Social Skills – Teamwork, Communication*

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

Puzzle: The Rook on a Number Grid

- You want to let rook move on the 4x4 board and help to reach the end position starting from left-top as shown in the figure. The Rook can only move the number of steps written on the cell.
- Start from the top-left square position and reach the bottom-right square. Jump in either direction (top, down, right or left) equal to the number on the square.
- Consider the example shown below. The 4x4 grid has numbers written in each square. The starting position (top-left square) has written number 2, you jumped two squares in the downwards direction.

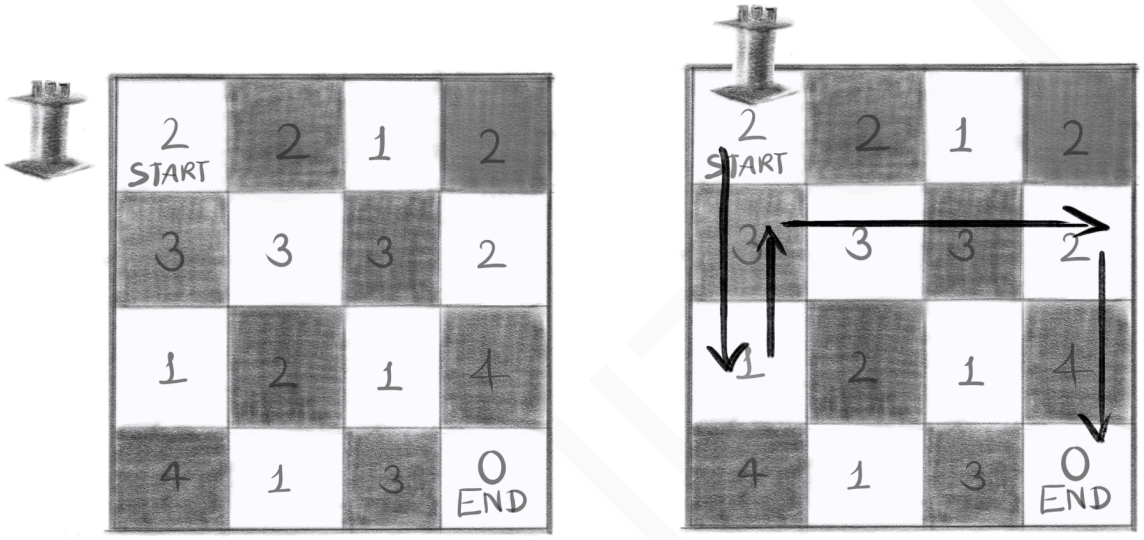


Figure: Help the rook reach the End, Example solution of the Puzzle

- **Not allowed - move diagonally**. The rook can move up-down-right-left, but not diagonally. Also, it **can not change the direction of steps intermediately**. As shown in the example, when you are at the start position you can not take 2 steps as 1 forward towards right and then 1 downwards. You have to move 2 whole steps in one direction either right or downwards in this case.

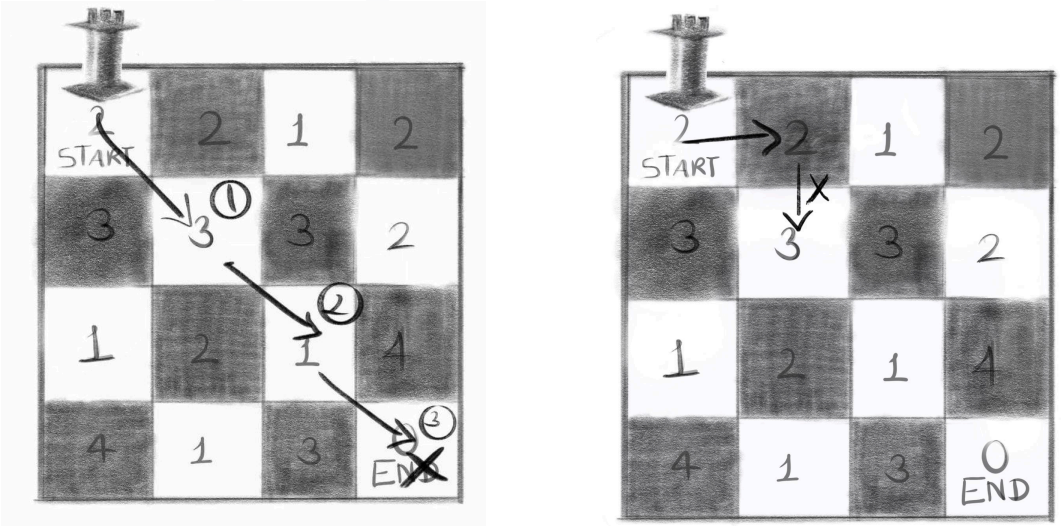


Figure: Not allowed - moving diagonally as shown above or take turn intermediatealy

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

Writing a Program

Computers follow instructions. To get our work done from computers, we need to tell them what to do. Unlike how we communicate, computers understand very few instructions. A programmer has to learn how to tell the computer what to do using these instructions.

---

*The list of instructions for computers is called **Program**.*

In this activity, we shall write a set of instructions for the robot/computer to reach from Start to End. There are lots of different programming languages a programmer can choose to write instructions. But we will be using a simple language that can be used without a computer. Instructions: Move forward (1 Step), Take right turn

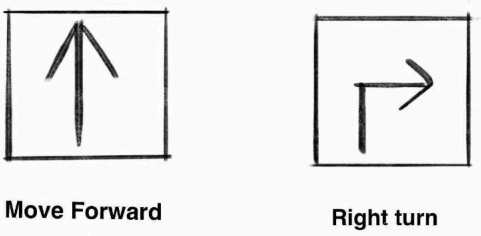


*Figure: Restricted steps in Number Maze Puzzle a.k.a **instructions***

One possible set of instructions (program) to solve the puzzle is shown below. The starting position is top-left square facing downwards. The Program:

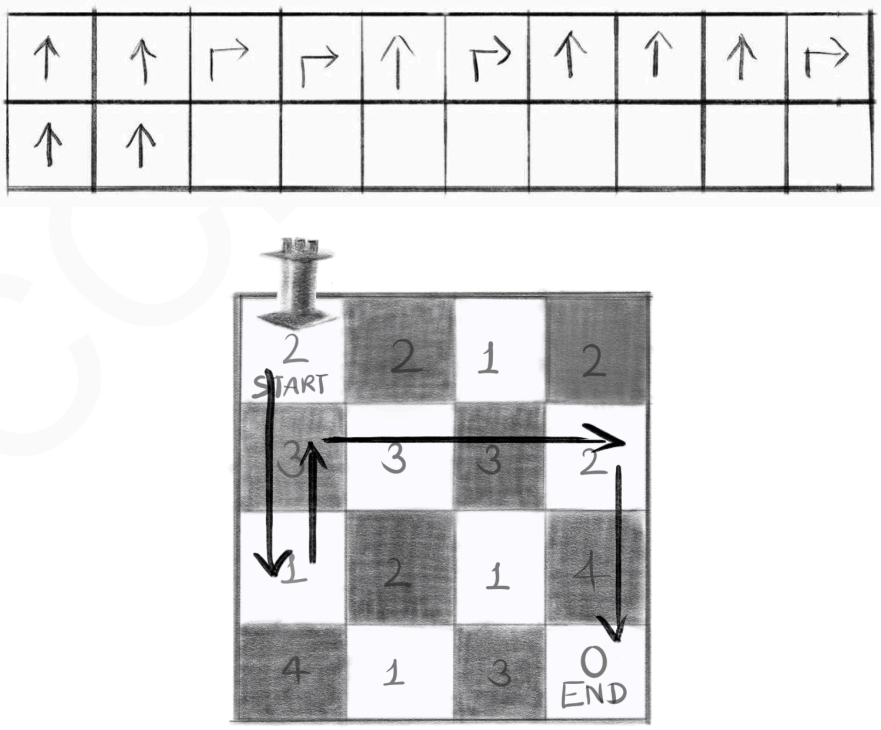


*Figure: **Program** (Set of Instructions) to reach the Destination in the Puzzle*

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

Note for Teachers: Classroom Set-Up

- Draw the Number-grid Puzzle and example on the board. Explain the puzzle with an example solution.
- Draw the number-grid shown in the image on a floor using chalk. The square tiles are great as they can serve as a grid. Invite students to try out the solutions by jumping over the grid.
- Once the puzzle is solved on pen-paper, write a program using only 3 types of instructions - Move Forward, Right Turn
- Make a group of 3 students - a programmer, a robot and a tester. The programmer will read the program loudly and communicate the instructions. The robot will follow the instructions over a puzzle grid on the foyer. The tester will check if the program is solving the puzzle or not.

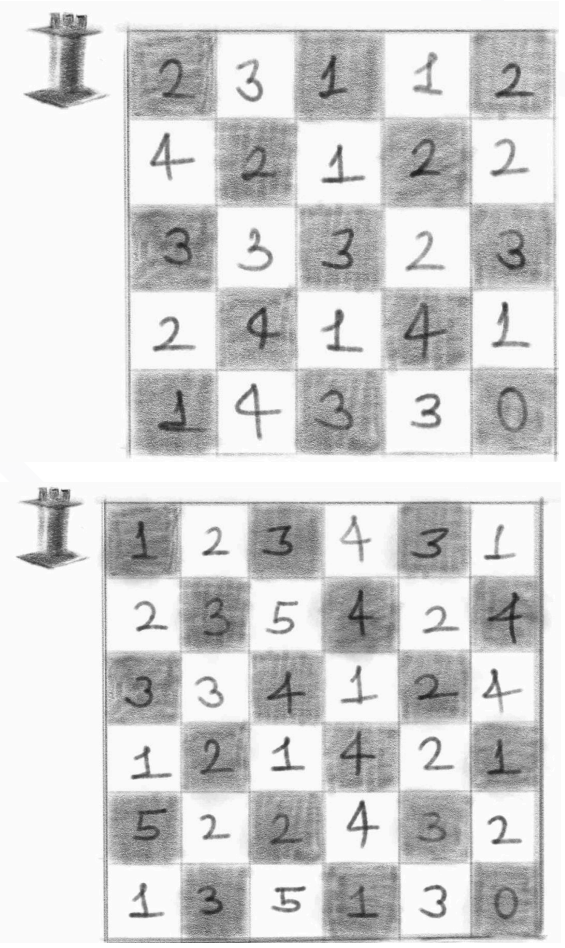


Figure: Number-grid puzzle exercises to try in classroom

We all are surrounded by Programs, not just in desktop computers or laptops, but in things like fitness devices, smartphones, alarm systems, TVs, watching machines and in many many more places. However in all such places the basics of programming is very similar - set of instructions to perform certain tasks. Sometimes it may seem like programming is some kind of special talent. But that's not true. Like all skills, programming is like learning through

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

practice, making mistakes, and learning from them. The most important part is to be able to write our thoughts in a set of instructions.

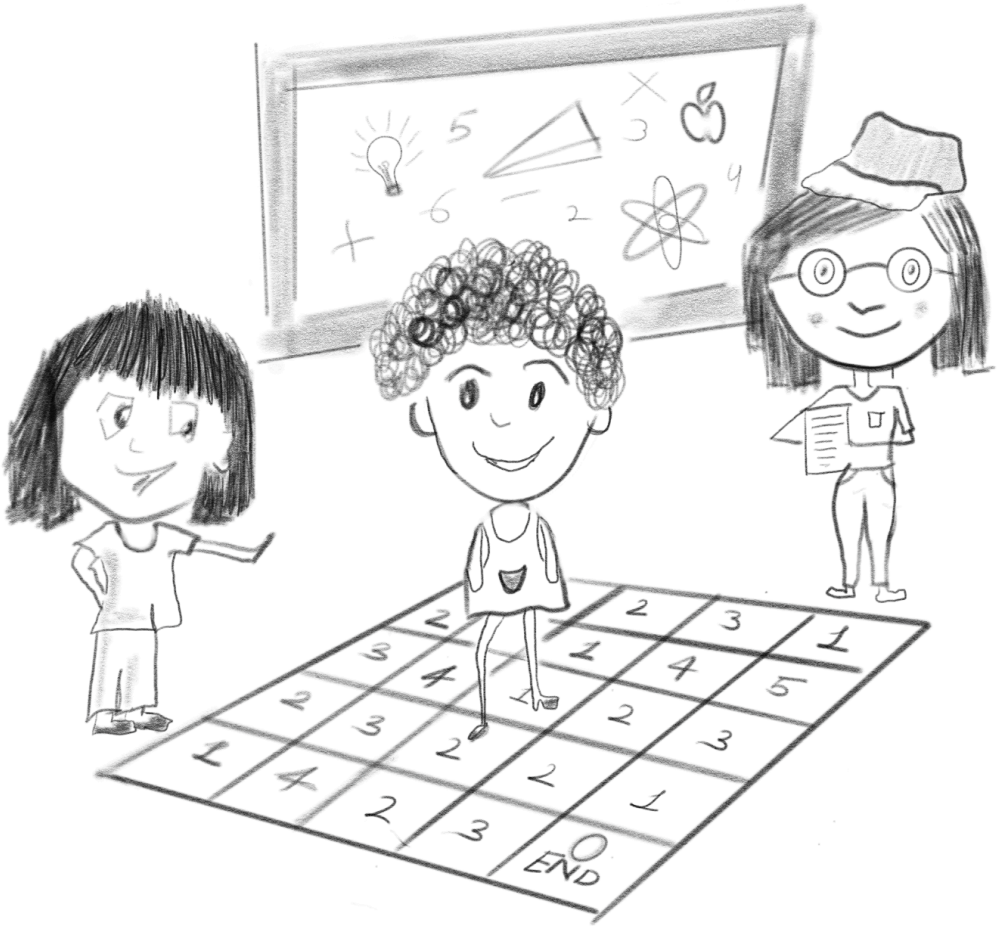


Figure: Classroom play inside the classroom,
3 pair of students - a programmer, a robot and a tester

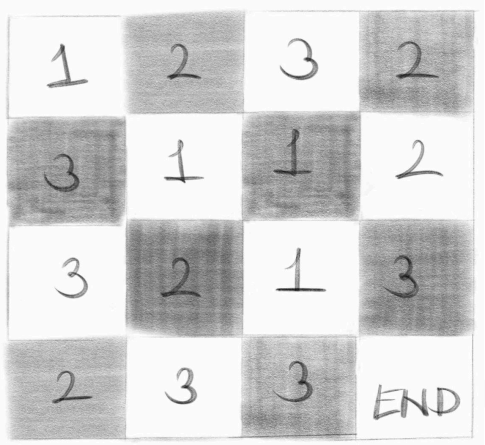
Writing a program involves thinking about how to get our task done from a computer. Here, we have explored a simple programming language with only two limited set of instructions (Move forward and Right Turn) to perform the solution of a given puzzle. The program is made because there is a problem to solve to help other people to improve their lives. The one who write the program using a programming language are called the programmer (or a software engineer) - they write the instructions to tell the computer what to do. To be sure that the program is working perfectly, someone needs to check it. They are called testers. In a simpler tasks/project usually the testing is done by the programmer only, but for larger applications/software testing is often separated out, partly to spread the workload, but also because sometimes the programmer might not be able to see its own mistake and user's perspective.

Being a programmer isn't only about knowing the particular instructions in a programming language, but the ability to think and put together these instructions in a way that the computer will do what is intended.

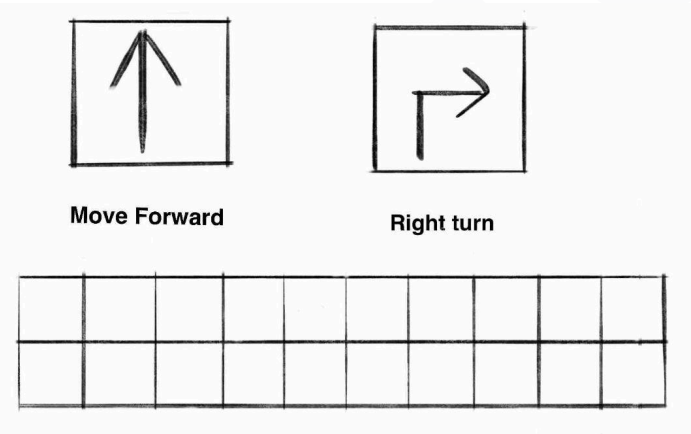
© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

Exercise

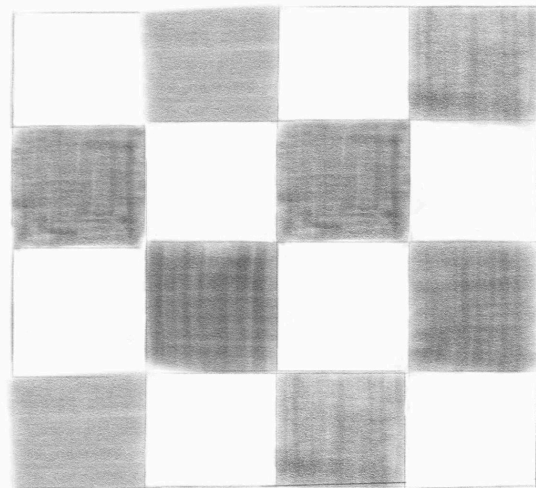
The Puzzle - Level 1 (4x4 Number Grid)



Write Program using only these two instructions: Move Forward & Right Turn

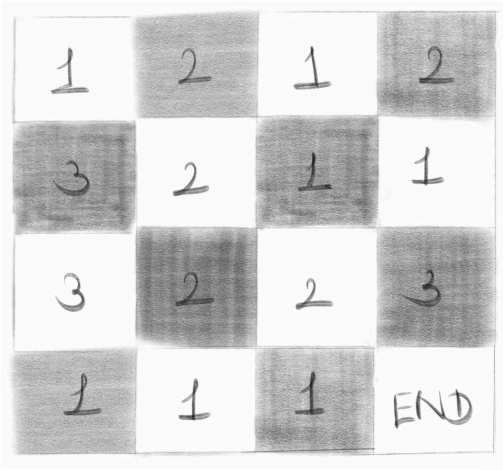


Test your program here (follow the instructions written above)

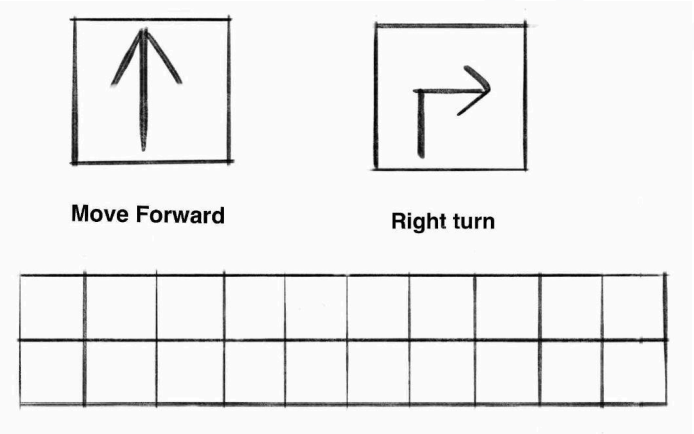


© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

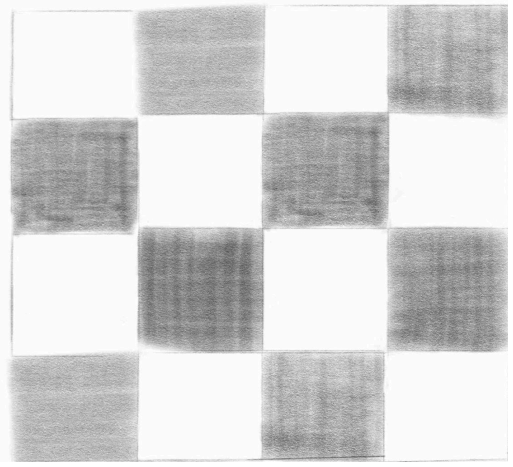
The Puzzle - Level 1 (4x4 Number Grid)



Write Program using only these two instructions: Move Forward & Right Turn

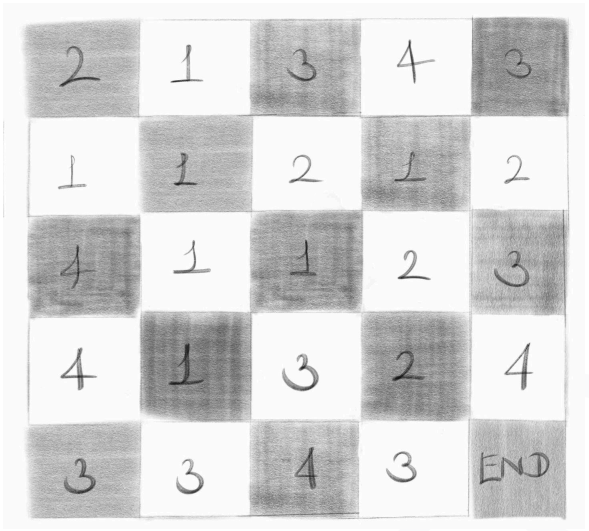


Test your program here (follow the instructions written above)

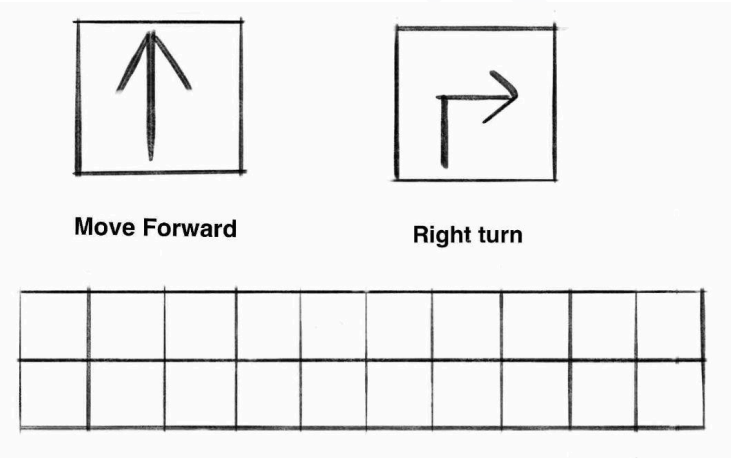


© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

The Puzzle - Level 2 (5x5 Number Grid)

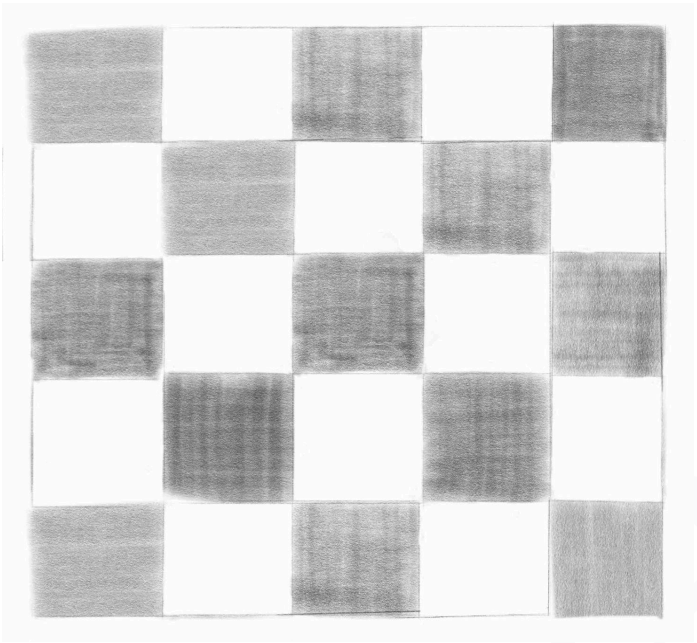


Write Program using only these two instructions: Move Forward & Right Turn



Test your program here (follow the instructions written above)

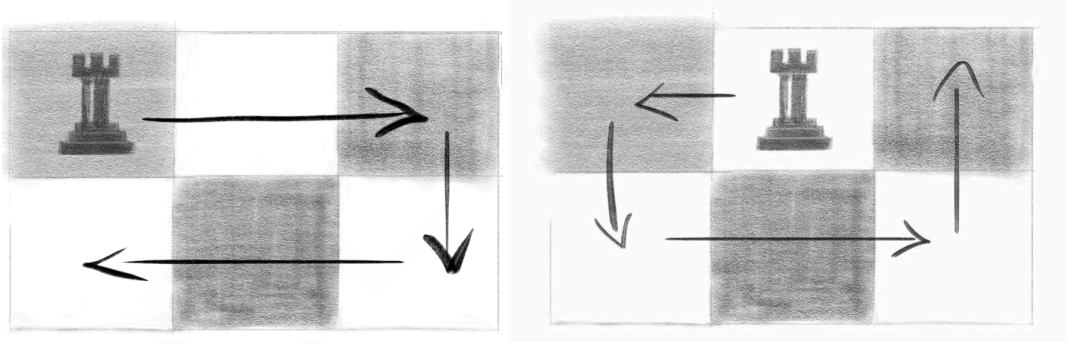
© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.



See and Solve Puzzles

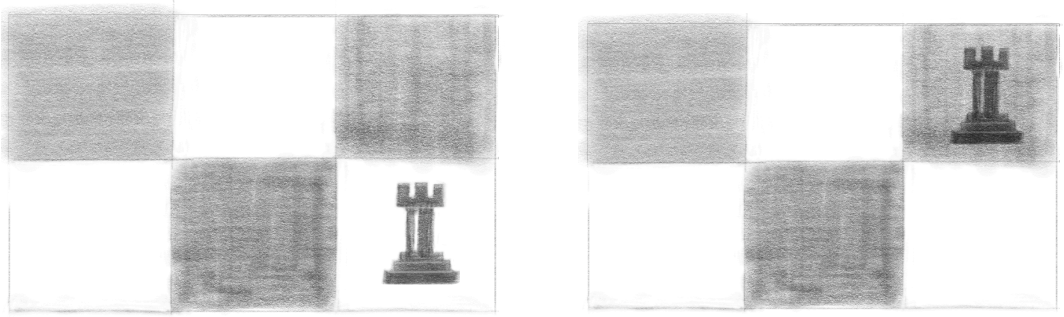
Rook on a Chess Board

2 x 3 Board Size

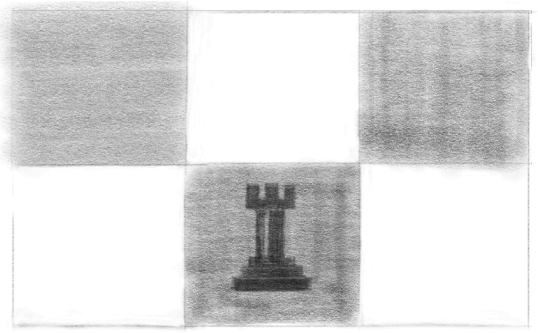


Example: Let the rook move on a board such that it covers all the squares, not repeating any square twice

Find the path starting from rook to cover all the squares !!



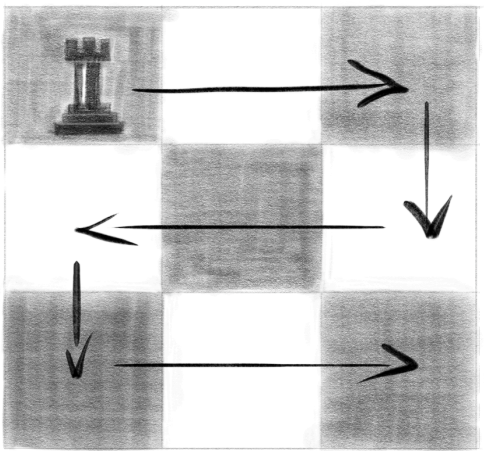
© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.



CCL - IITGN

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

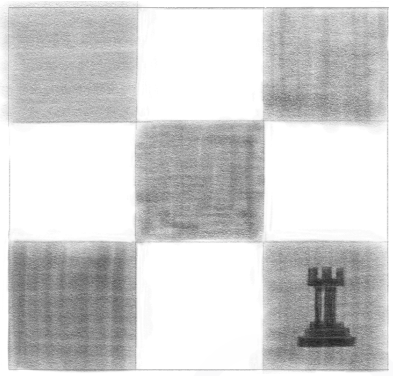
3 x 3 Board Size



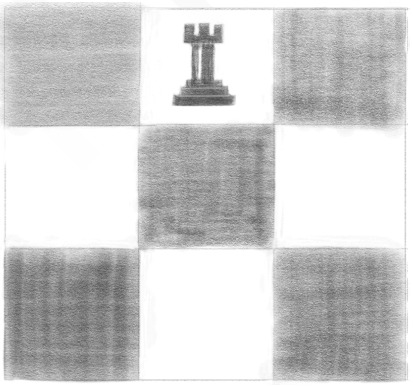
Path Possible? *Yes*

Example: Same rook puzzle but on a 3x3 chess board.
See for a given rook position is it possible to find a path that covers all the cell ?

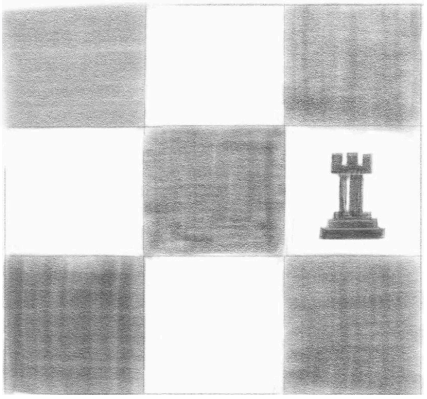
Possible to find the path covering all the squares given rook's position?



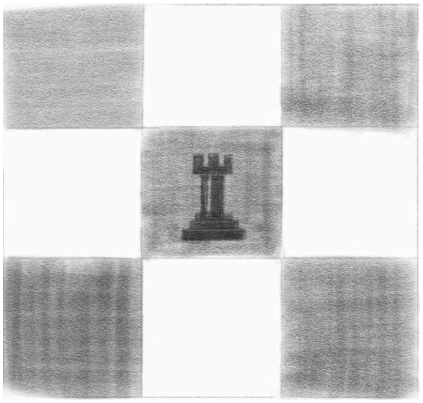
Path Possible? _____



Path Possible? _____



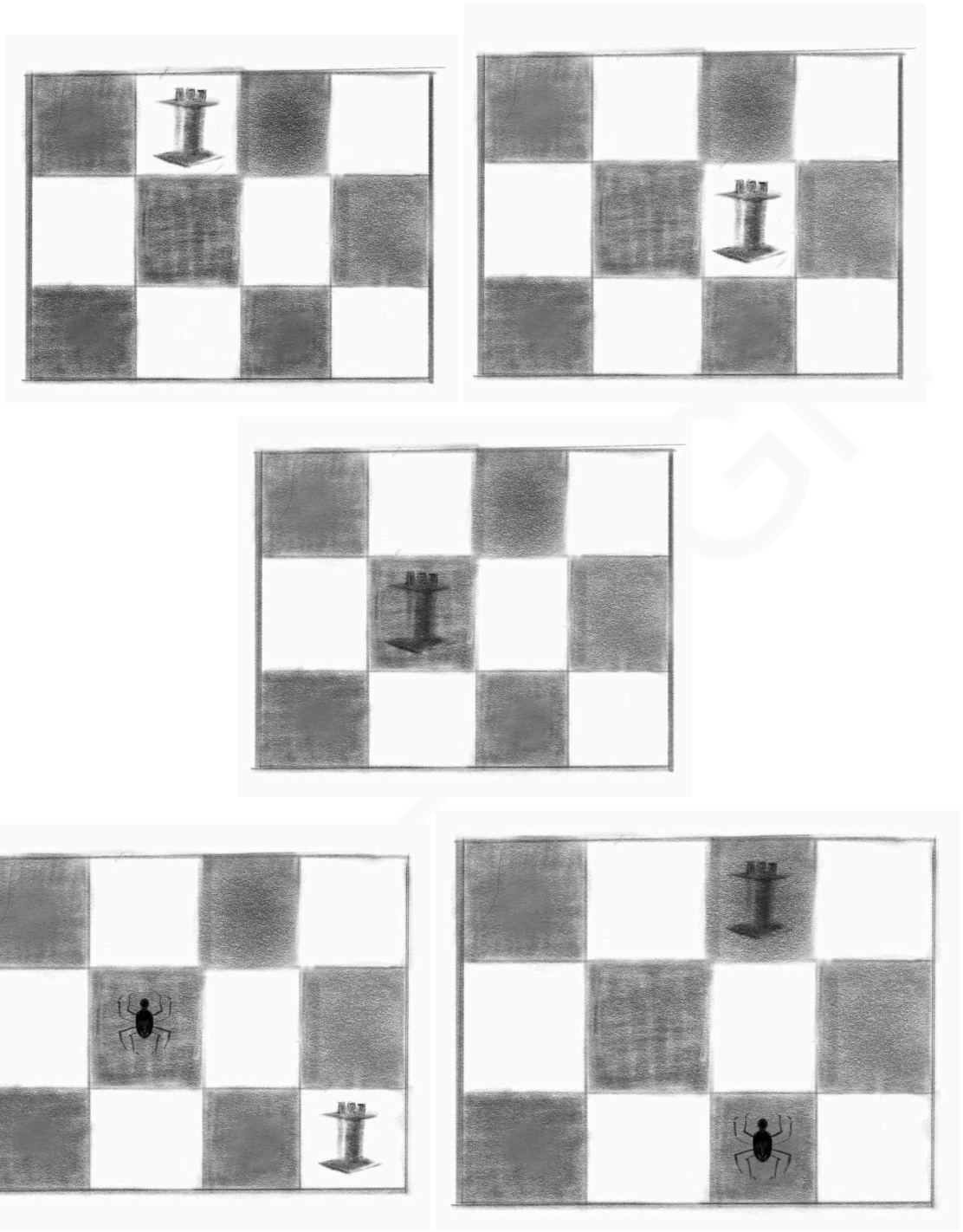
Path Possible? _____



Path Possible? _____

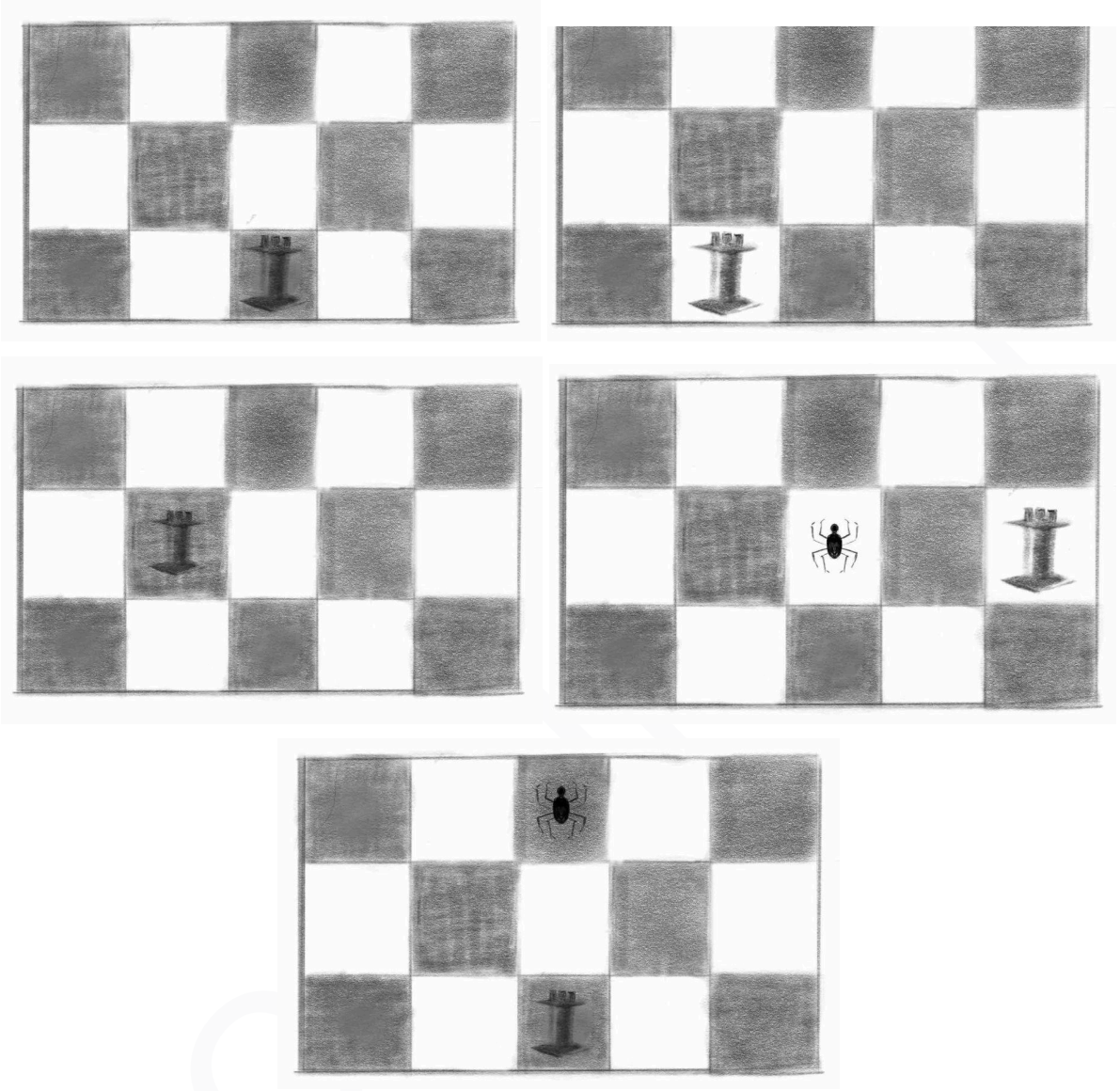
© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

3 x 4 Board Size



© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

3 x 5 Board Size



Think:

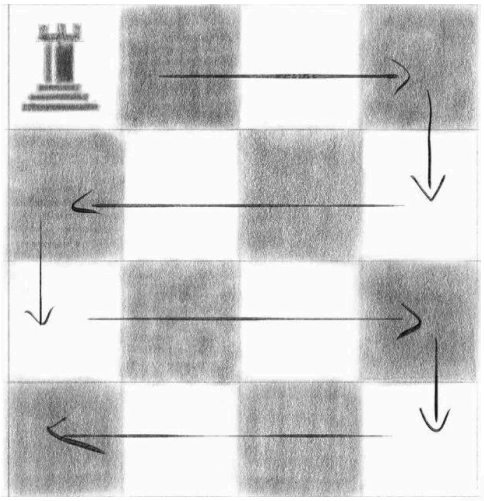
Why for some rook positions we are not able to find the path covering all the squares?

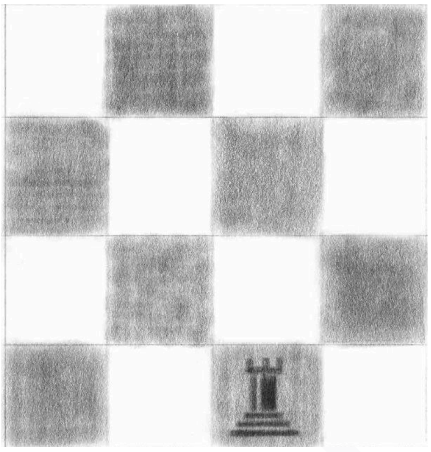
By just looking at board size and rook's position can we tell whether its possible to find the path or not ??

Hint:
In the checker board, the path will have an alternative white and black square. You can never travel from one white square to another white

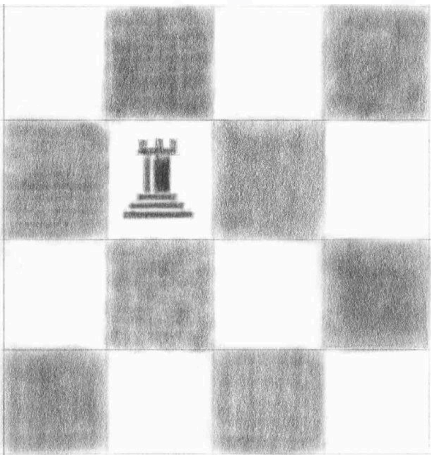
© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

4 x 4 Board Size

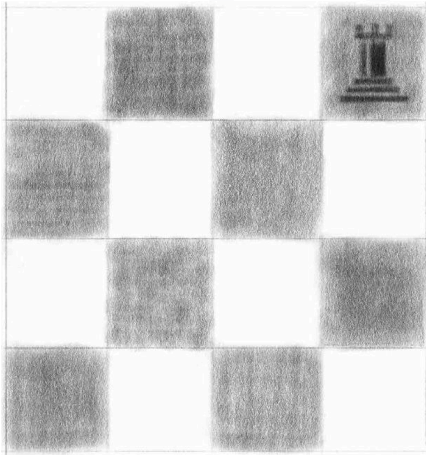




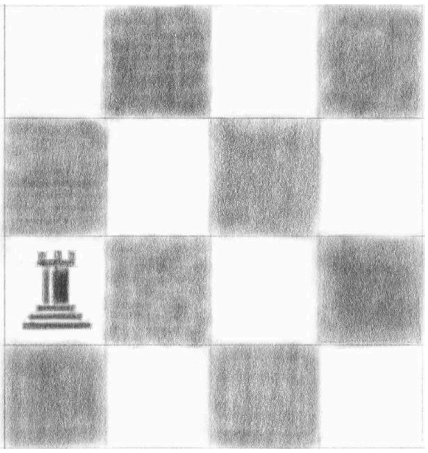
Path Possible? _____



Path Possible? _____

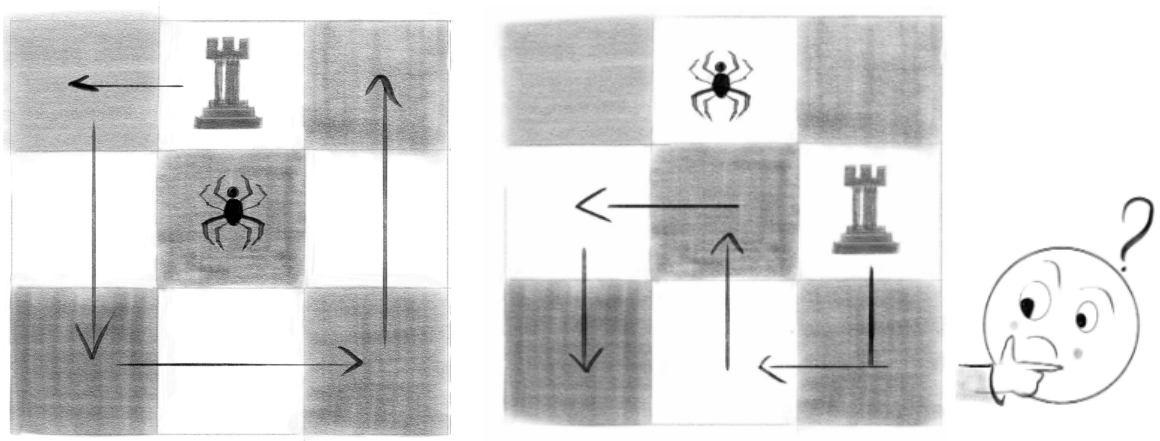


Path Possible? _____

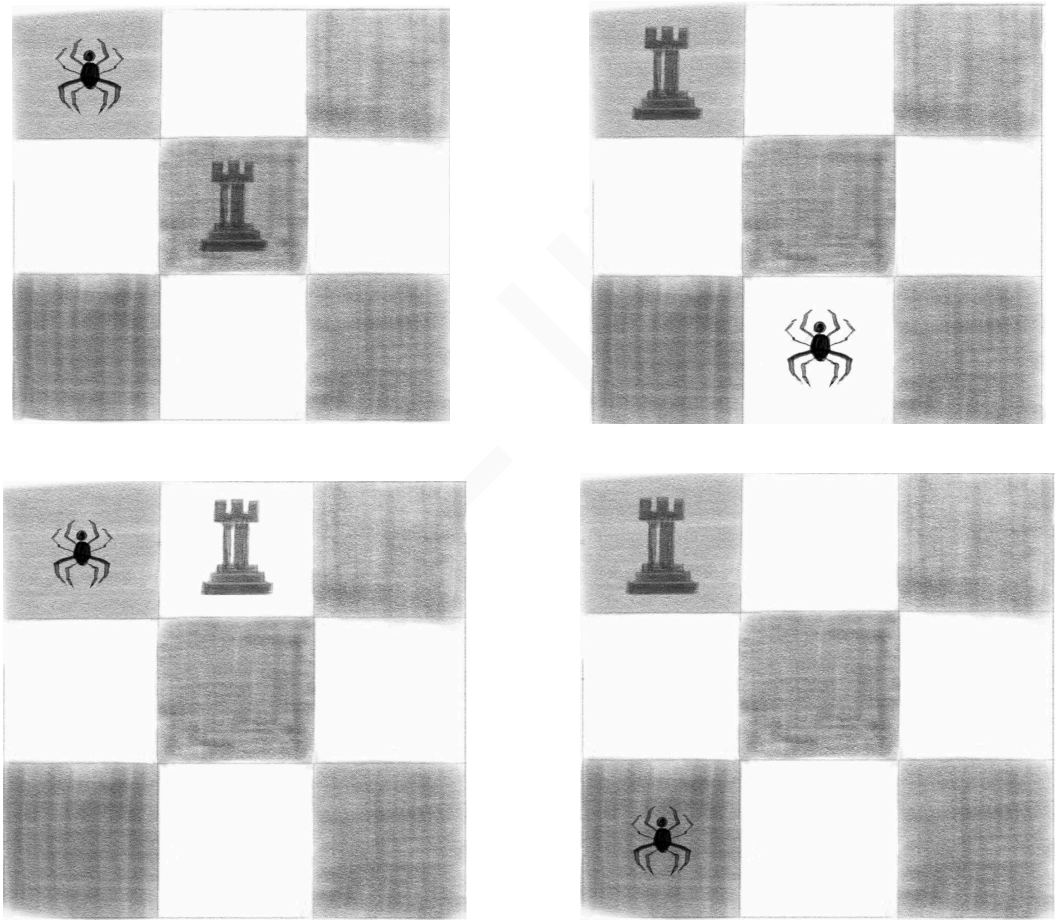


Path Possible? _____

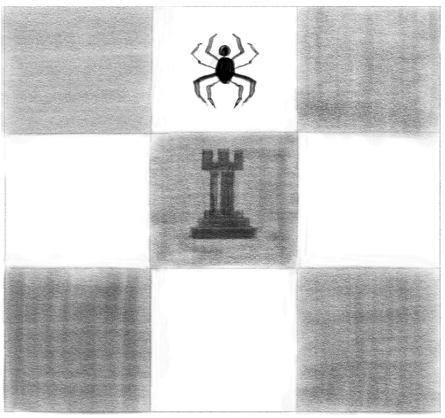
© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

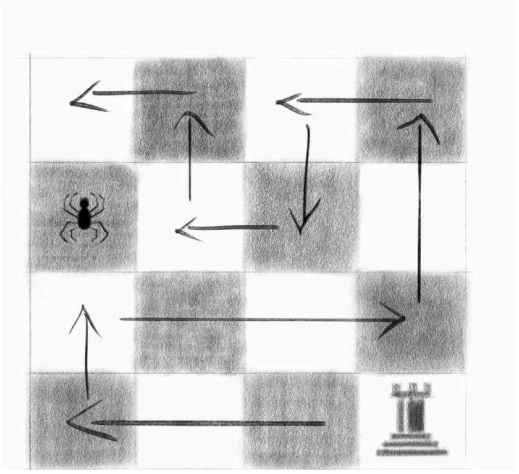


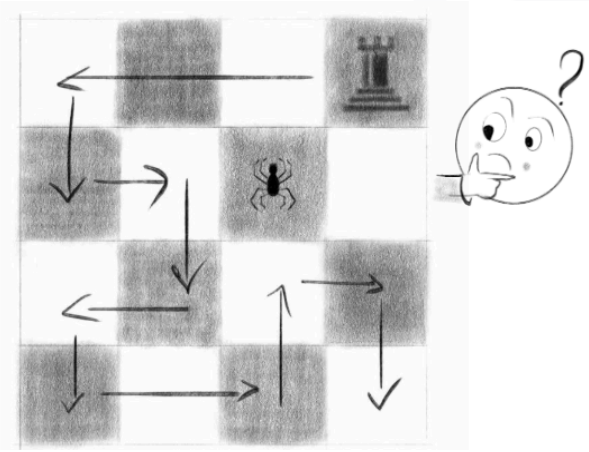
Example: Spider on a Chess board, cover all the cells except the one with spider !!



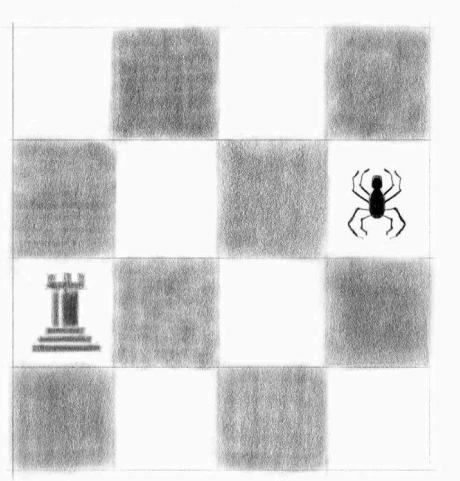
© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

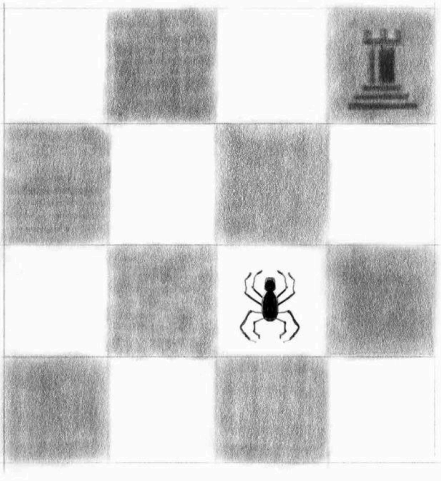


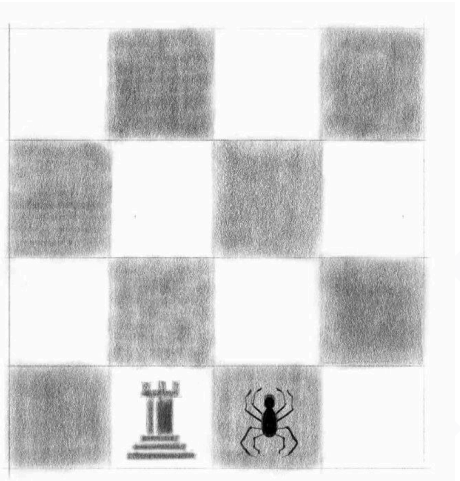


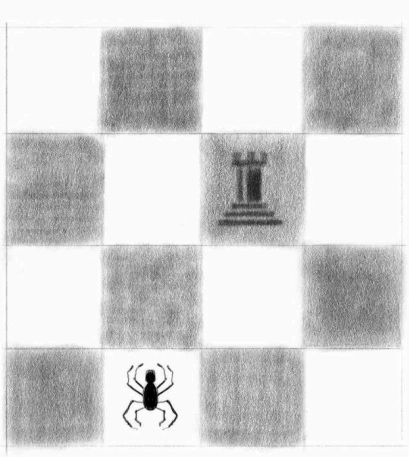


© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.



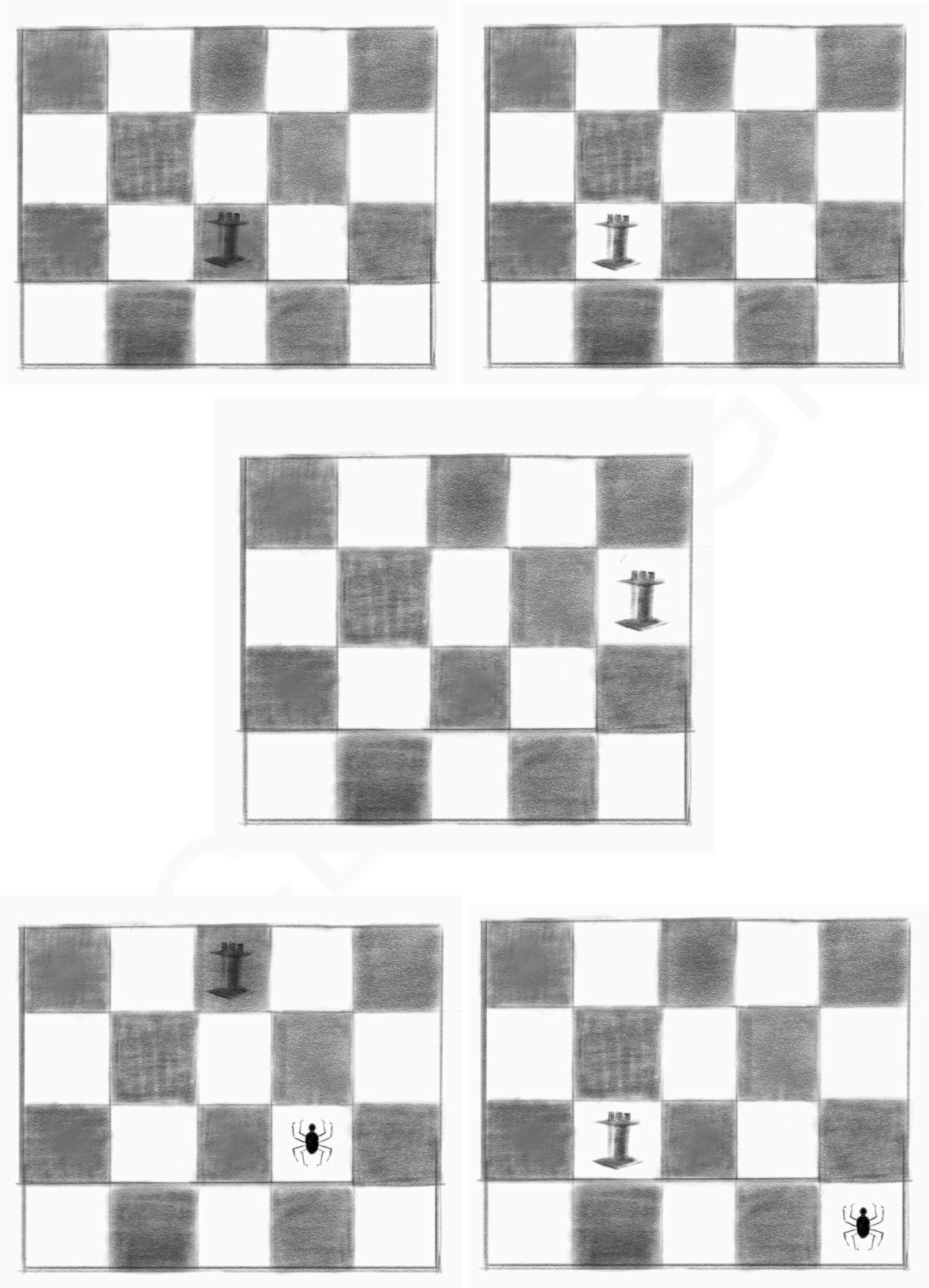






© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

4 x 5 Board Size



© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.

1. **Fill the following by the number of boxes from which you have passed.**

Puzzle	No. of black squares	No. of white square
Puzzle a	3	2
Puzzle b		
Puzzle c		
Puzzle d		
Puzzle e		
Puzzle f		
Puzzle g		
Puzzle h		
Puzzle i		

2. Write a sequence of the path you followed while solving it.
   For Ex.1 WBWBW

   W for white squares
   B for black square

CCL - IITGN

© 2026 [Center for Creative Learning, IIT Gandhinagar]. All rights reserved. For educational use only.